

Course No. CSSE 601

Distributed Systems

Course No. CSSE-601

System Architecture

Objective

- System Architecture
 - Centralized Organizations
 - Simple Client-Server Architecture
 - Multitiered Architecture
 - Decentralized Organizations
 - Peer-to-Peer Systems
 - Structured Peer-to-Peer Systems
 - Unstructured Peer-to-Peer Systems
- Decentralized Organizations
 - Structured Peer-to-Peer Systems
 - Unstructured Peer-to-Peer Systems
 - Hierarchically Organized Peer-to-Peer Networks
- Hybrid Architectures
 - Edge-Server Systems
 - Collaborative Distributed Systems

System Architecture

- Deciding on software components, their interaction, and their placement leads to an instance of a software architecture, also known as a system architecture.

System Architecture

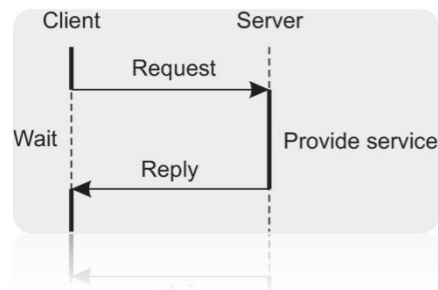


Centralized Organizations

- **Simple Client-Server Architecture:** To design a distributed system, processes are divided into two groups: Like Server and Client.
- A server is a process implementing a specific service.
- A client is a process that requests a service from a server by sending it a request and subsequently waiting for the server's reply.
- This environment is also known as request-reply behavior.

Centralized Organizations

- Simple Client-Server Architecture

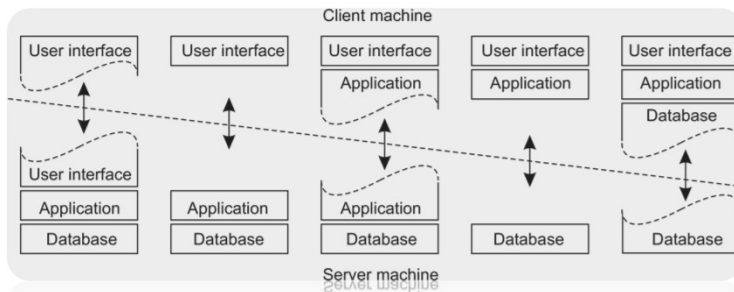


Centralized Organizations

- **Multitiered Architecture:** Many Distributed applications are divided into the three layers (1) user interface layer, (2) processing layer, and (3) data layer.

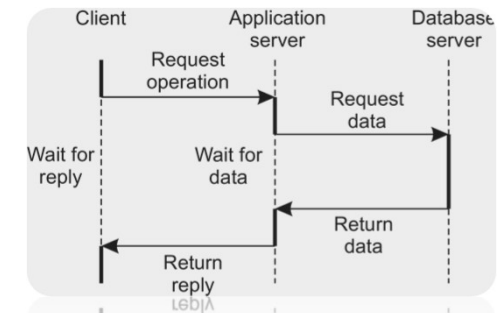
Centralized Organizations

• Multitiered Architecture



Centralized Organizations

• Multitiered Architecture



Decentralized Organizations

- It also called peer-to-peer systems.
- Multitiered client-server architectures are a direct consequence of dividing distributed applications into a user interface, processing components, and data-management components.
- In many business environments, distributed processing is equivalent to organizing a client-server application as a multitiered architecture. This type of distribution as **Vertical Distribution**.
- The distribution of the clients and the servers that counts, which we refer to as **horizontal distribution**.

Decentralized Organizations

- Peer-to-Peer Systems:
- Peer-to-peer architectures evolve around the question how to organize the processes in an overlay network.
- A node may not be able to communicate directly with an arbitrary other node, but is required to send messages through the available communication channels.
- Two types of overlay networks exist: Structured and Non Structured.

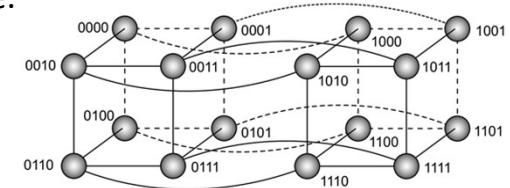
Decentralized Organizations

• Structured Peer-to-Peer Systems:

- In a structured peer-to-peer system the nodes are organized in an deterministic topology.
- Like Ring, Binary Tree, Grid etc.
- $\text{key}(\text{data item}) = \text{hash}(\text{data item's value})$.

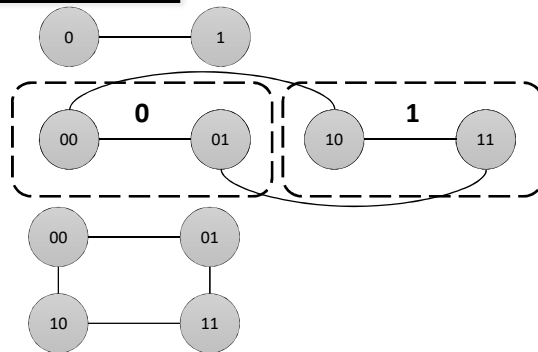
Decentralized Organizations

- Structured Peer-to-Peer Systems: simple peer-to-peer system with a fixed number of nodes, organized into a hypercube. A hypercube is an n-dimensional cube.



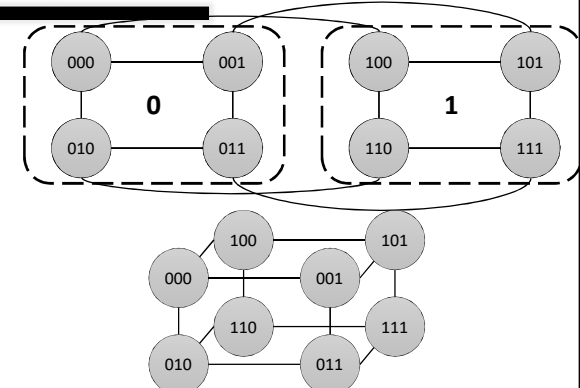
Decentralized Organizations

- Creating Hyper Cube



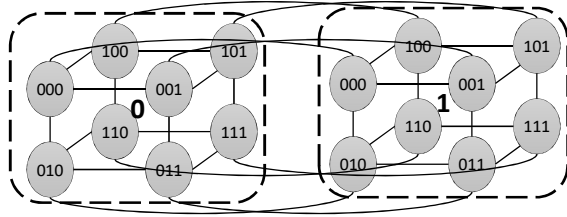
Decentralized Organizations

- Creating Hyper Cube



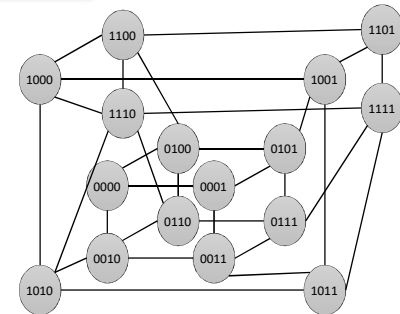
Decentralized Organizations

- Creating Hyper Cube



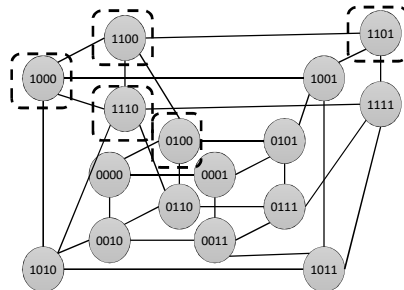
Decentralized Organizations

- Creating Hyper Cube



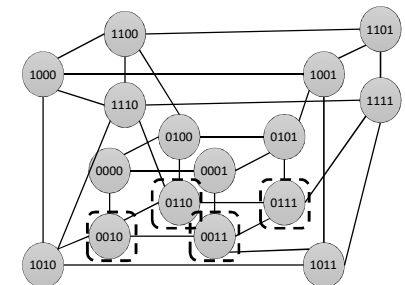
Decentralized Organizations

- What nodes connect to node 12?
- Node 12 = $(1100)_2$
- Apply the XOR operation with a "1" digit in each of the four positions.
- $1100 \text{ XOR } 0001 = 1101 (13)_{10}$
- $1100 \text{ XOR } 0010 = 1110 (14)_{10}$
- $1100 \text{ XOR } 0100 = 1000 (8)_{10}$
- $1100 \text{ XOR } 1000 = 0100 (4)_{10}$
- Answer: **4, 8, 13, 14**



Decentralized Organizations

- Find route from nodes 2 to node 7?
- $2 = (0010)_2$
- $7 = (0111)_2$
- Apply XOR operation.
- $0010 \text{ XOR } 0111 = 0101$
- $0010 - x1xx = 0110 (6)_{10}$
- $0010 - xxx1 = 0011 (3)_{10}$
- Path
 - **2 -> 3 -> 7**
 - **2 -> 6 -> 7**



Decentralized Organizations

- **Unstructured Peer-to-Peer Systems:** In unstructured peer-to-peer system each node maintains an ad hoc list of neighbors.
- A node joins it often contacts a well-known node to obtain a starting list of other peers in the system.

Decentralized Organizations

- **Flooding:** Simple passes a **request** for a data **item to all its neighbors**. A **request** will be ignored when its **receiving node**, had seen it before. (e.g. TTL, high, low)
- **Random Walks:** It can simply try to find a data item by asking a **randomly chosen neighbor**, it **forwards** the **request** to one of its randomly chosen neighbors and so on.

Policy-based search methods

Decentralized Organizations

- Deploying random walks is not only effective, it can also be much more efficient in comparison to

$$\mathbb{P}[k] = \frac{r}{N} \left(1 - \frac{r}{N}\right)^{k-1}.$$

$$S = \sum_{k=1}^N k \cdot \mathbb{P}[k] = \sum_{k=1}^N k \cdot \frac{r}{N} \left(1 - \frac{r}{N}\right)^{k-1} \approx N/r \text{ for } 1 \ll r \leq N.$$

- Compare this to flooding.

$$R(k) = d(d-1)^{k-1}$$

Decentralized Organizations

- Suppose r or $d = 10$, $N=10,0000$ and $k= 4$

Random

$$P[k] = r/N (1 - r/N)^{k-1}$$

$$P[4] = 10/10000 (1-10/10000)^{4-1}$$

$$P[4] = 0.001(1-0.001)^3$$

$$P[4] = 0.000997$$

Flooding

$$R[k] = d \cdot (d-1)^{k-1}$$

$$R[4] = 10 (10-1)^{4-1}$$

$$R[4] = 10 (9)^3$$

$$R[4] = 7290$$

Convert into Probability

$$R[4] = 7290/10000 = 0.729$$

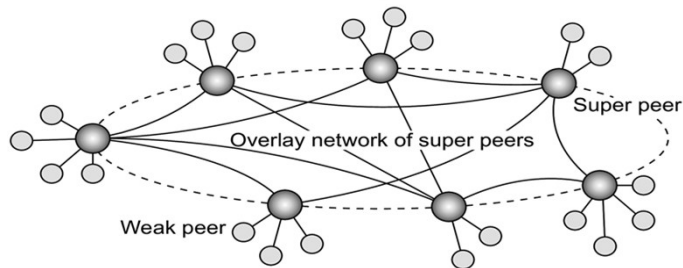
Decentralized Organizations

- **Problem:** There is no deterministic way of routing a lookup request to a specific data item, essentially the only technique a node can resort to is searching for the request by means of flooding or randomly walking through the network.

Decentralized Organizations

- **Hierarchically Organized Peer-to-Peer Networks.**
- **Content Delivery Network (CDN):** Nodes may offer storage for hosting copies of web documents allowing web clients to access pages nearby.
- **Broker:** Collects data on resource usage and availability for a number of nodes that are in each other's proximity.
- **Super Peers:** Node that maintaining an index or acting as a broker.
- **Week Peers:** is connected as a client to a super peer.

Decentralized Organizations



- Selection of super node by using **Leader-Election Problem**

Decentralized Organizations

- The Skype network is organized very similar to what is shown in *Figure* with one difference, there is an additional centralized **Skype Login Server**.
- There are a number of default Skype super peers, the address of each of these default super peers is hard-coded in Skype software.
- Weak peer has a local list of addresses of reachable super peers, (host cache). If none of the cached super peer reachable, it tries to connect to one of the default super peers.

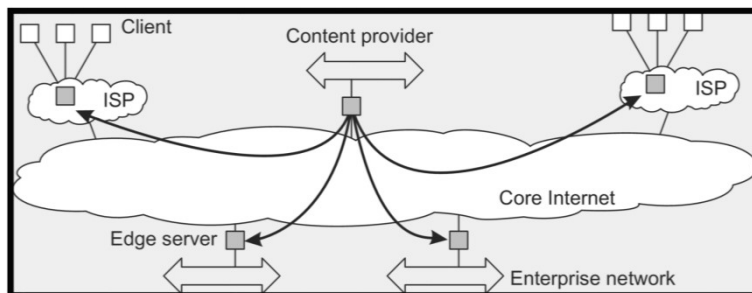
Hybrid Architectures

- Many distributed systems combine architectural features, in which client-server solutions are combined with decentralized architectures.

Hybrid Architectures

- **Edge-Server Systems:** these systems are deployed on the Internet where servers are placed “at the edge” of the network.
- The edge is formed by the boundary between enterprise networks and the actual Internet.

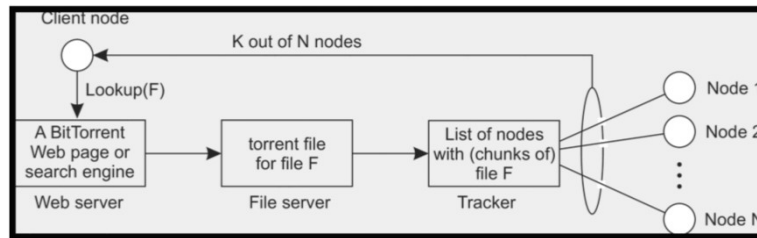
Hybrid Architectures



Hybrid Architectures

- **Collaborative Distributed Systems:** Hybrid structures are especially deployed in collaborative distributed systems.
- A traditional client-server scheme is deployed. Once a node has joined the system, it can use a fully decentralized scheme for collaboration. (E.g. BitTorrent)

Hybrid Architectures



Hybrid Architectures

- **Advanced BitTorrent:** The downloading of a file can be described in term of three phases
- **Bootstrap:** It is in a position to start trading blocks. It will have to wait until new neighbors are added to the set, who may have pieces to exchange.
- **Trading Phase:** the downloading is highly efficient as enough nodes can be paired to exchange data.
- **Last Download Phase:** when set has dropped to ZERO, it depends on newly arriving peers, to get the last missing piece.

Query Regarding Lecture

Email: dr. m.naseem622@gmail.com

Thanks

Email: dr. m.naseem622@gmail.com